

# **DRAEM - ICCV 2021**

A Discriminatively Trained Reconstruction Embedding for Surface Anomaly Detection – Vitjan Zavrtanik, Matej Kristan, Danijel Skočaj

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<sup>1</sup>Friedrich-Alexander-Universität Erlangen-Nürnberg, Technische Fakultät

January 11, 2026

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- Data collection gets even harder when pixel-accurate localization is required
- Therefore, methods that only rely on normal data are preferred

# Problem Statement

## Current Situation and Related Work

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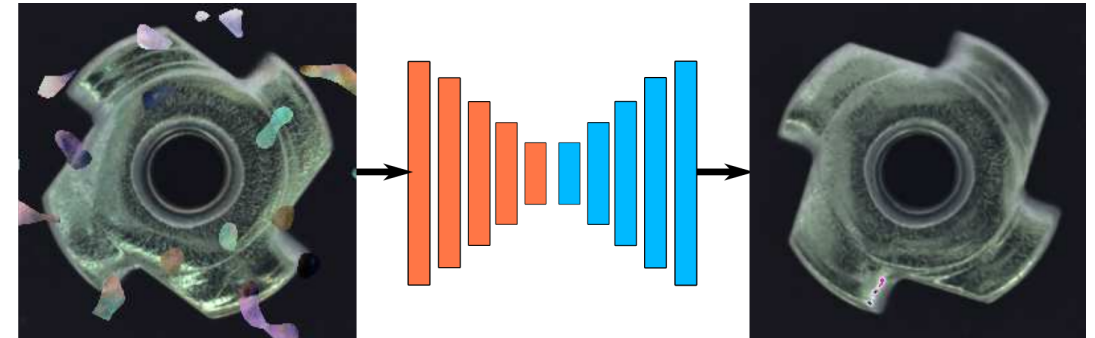
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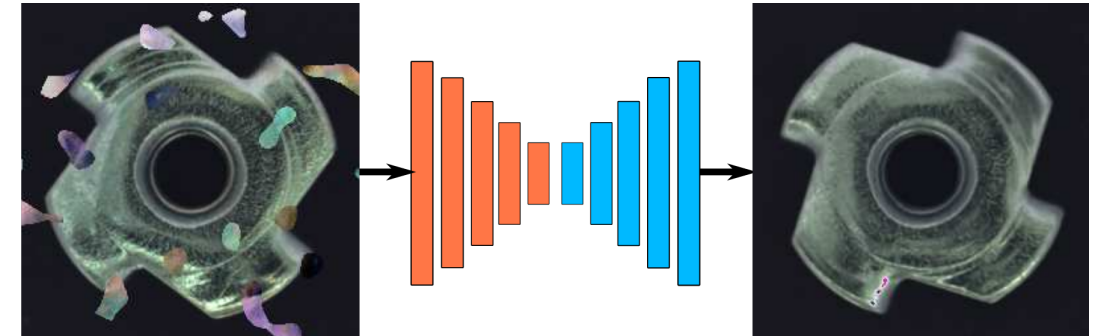
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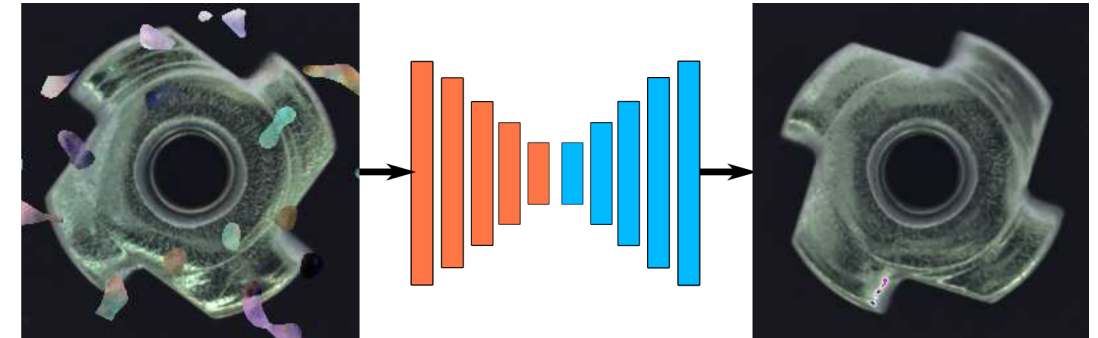
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  2. **Discriminative Methods** detect distribution deviations on (deep) features (e.g. from CNNs)
    - Often suffer from blurry or coarse segmentation
    - Overfit to synthetic anomalies

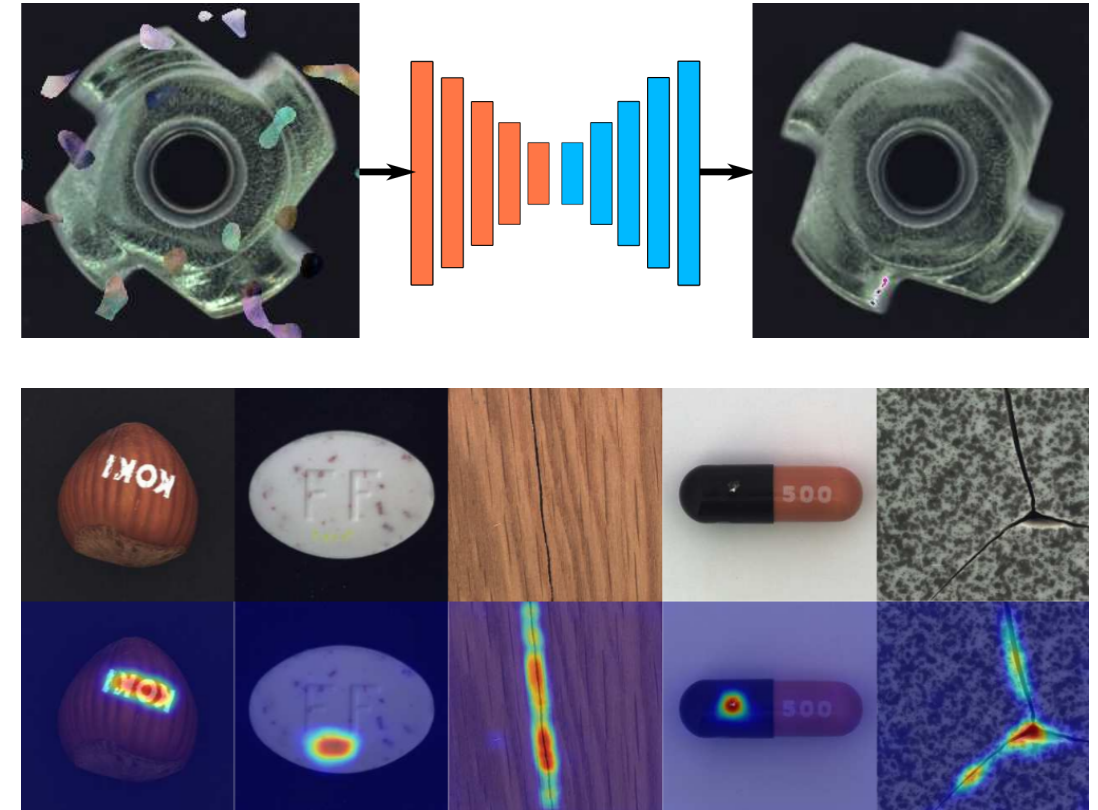
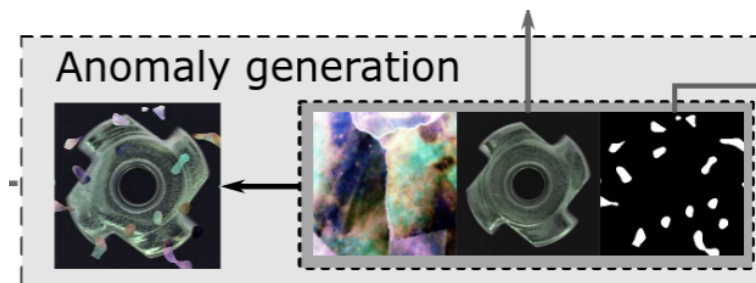


Figure: Blurry segmentation maps from discriminative methods.

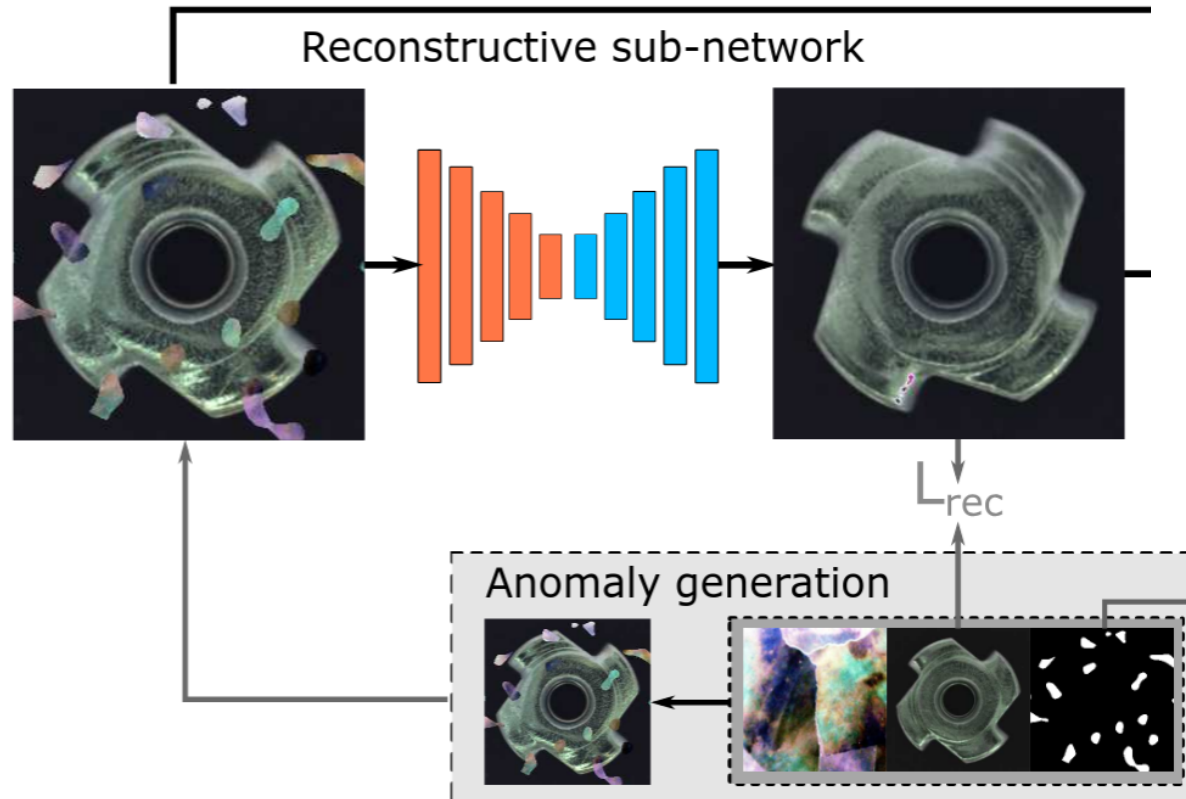
# Framework Overview

## How it Works



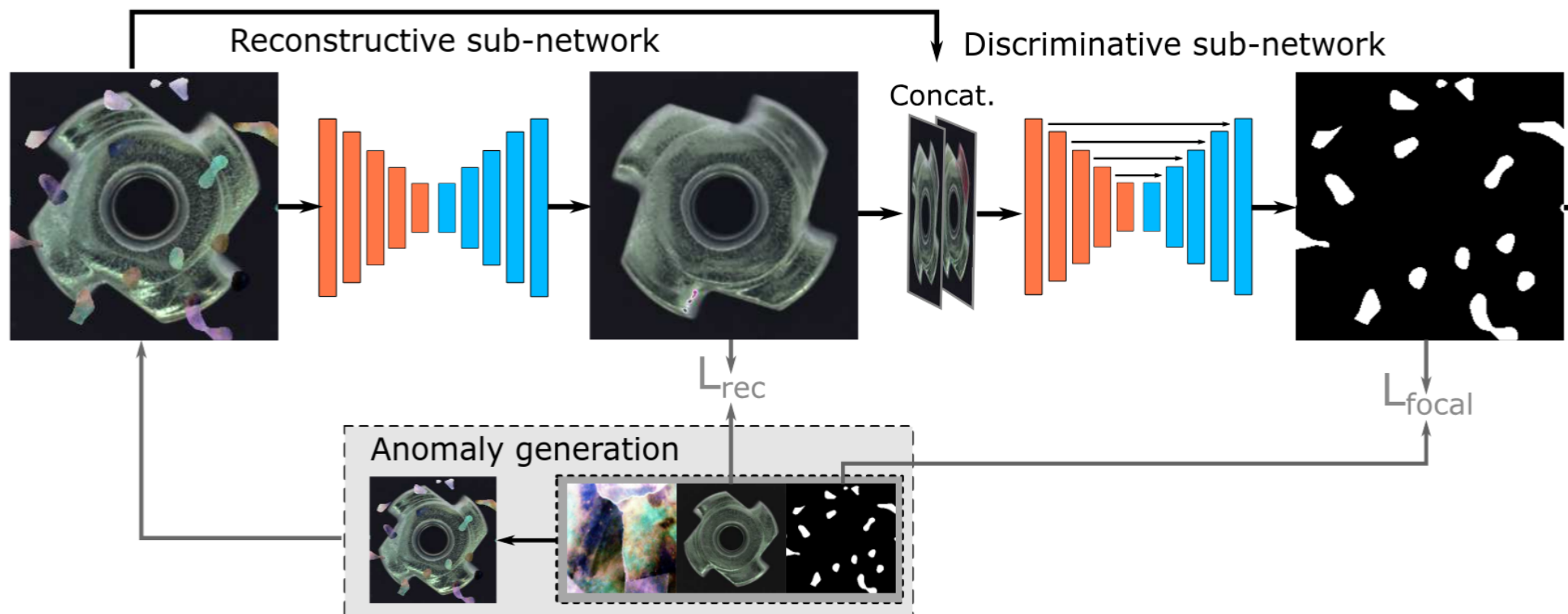
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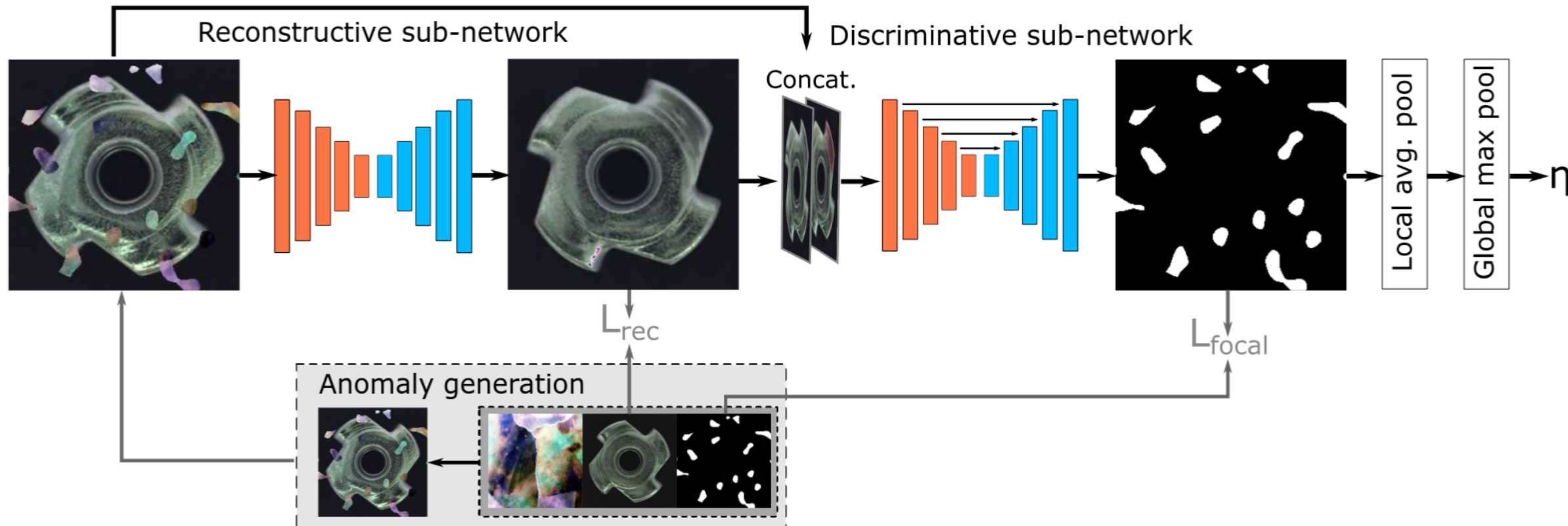
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# Performance Comparison

## DRÆM vs State-of-the-Art

| Method | Image AUROC  | Pixel AUROC  | Pixel AP     |
|--------|--------------|--------------|--------------|
| US     | 87.7%        | 93.9%        | 45.5%        |
| RIAD   | 91.7%        | 94.2%        | 48.2%        |
| PaDim  | <u>95.5%</u> | <b>97.4%</b> | <u>55.5%</u> |
| DRÆM   | <b>98.0%</b> | <u>97.3%</u> | <b>68.4%</b> |

Table: Results on the MVTec dataset.

# Performance on DAGM Dataset

## Unsupervised vs. Supervised Methods

| Method              | AUROC       | TPR         | TNR         | CA          |
|---------------------|-------------|-------------|-------------|-------------|
| <b>Unsupervised</b> |             |             |             |             |
| US                  | 72.5        | 72.6        | 65.3        | 66.2        |
| RIAD                | 78.6        | 79.2        | 69.1        | 70.4        |
| PaDim               | 95.0        | 83.3        | 97.5        | 95.7        |
| DRÆM                | <b>99.0</b> | <b>96.5</b> | <b>99.4</b> | <b>98.5</b> |

**Table:** DRÆM on DAGM dataset: outperforming unsupervised methods and on par with fully supervised ones.

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| Racki et al.        | 99.6        | 99.9        | 99.5        | —           |
| Lin et al.          | 99.0        | 99.4        | 99.9        | —           |
| Bozic et al.        | <b>100</b>  | <b>100</b>  | <b>100</b>  | <b>100</b>  |

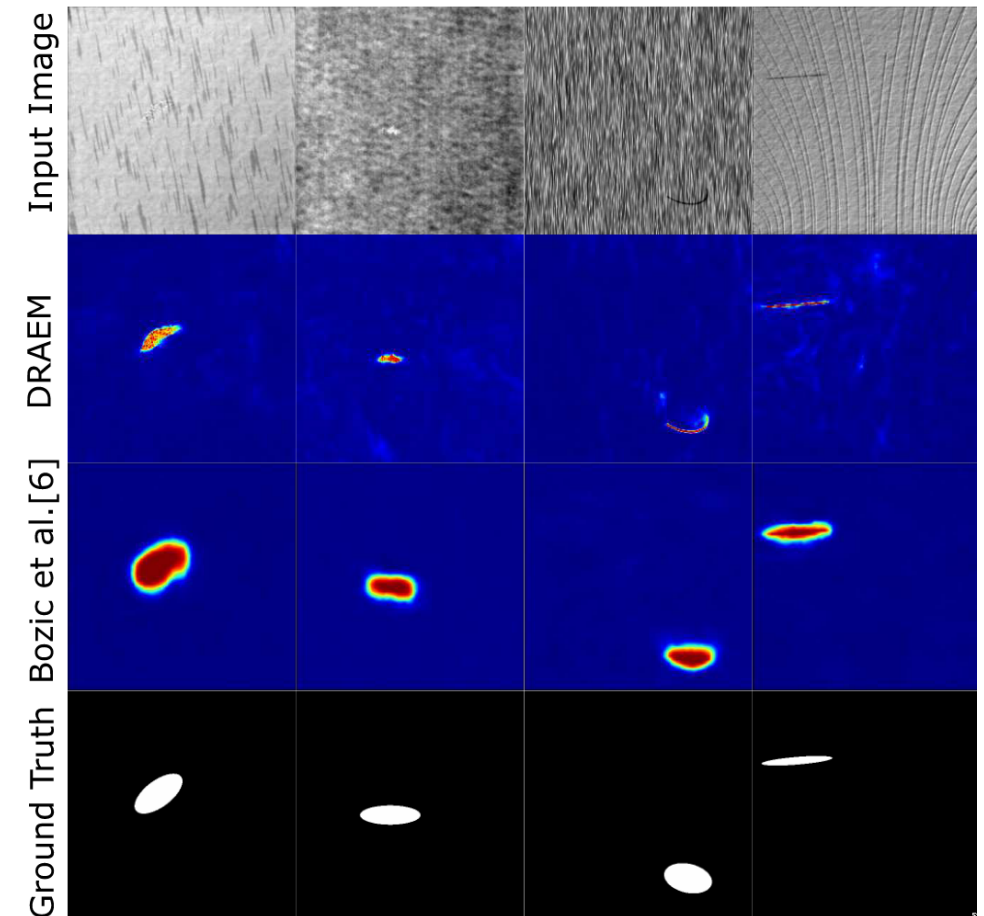
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**Table:** DRÆM on DAGM dataset: outperforming unsupervised methods and on par with fully supervised ones.



**Figure:** DRÆM vs supervised SOTA (Bozic et al.) on the DAGM dataset.

# Key Findings & Application & Conclusion

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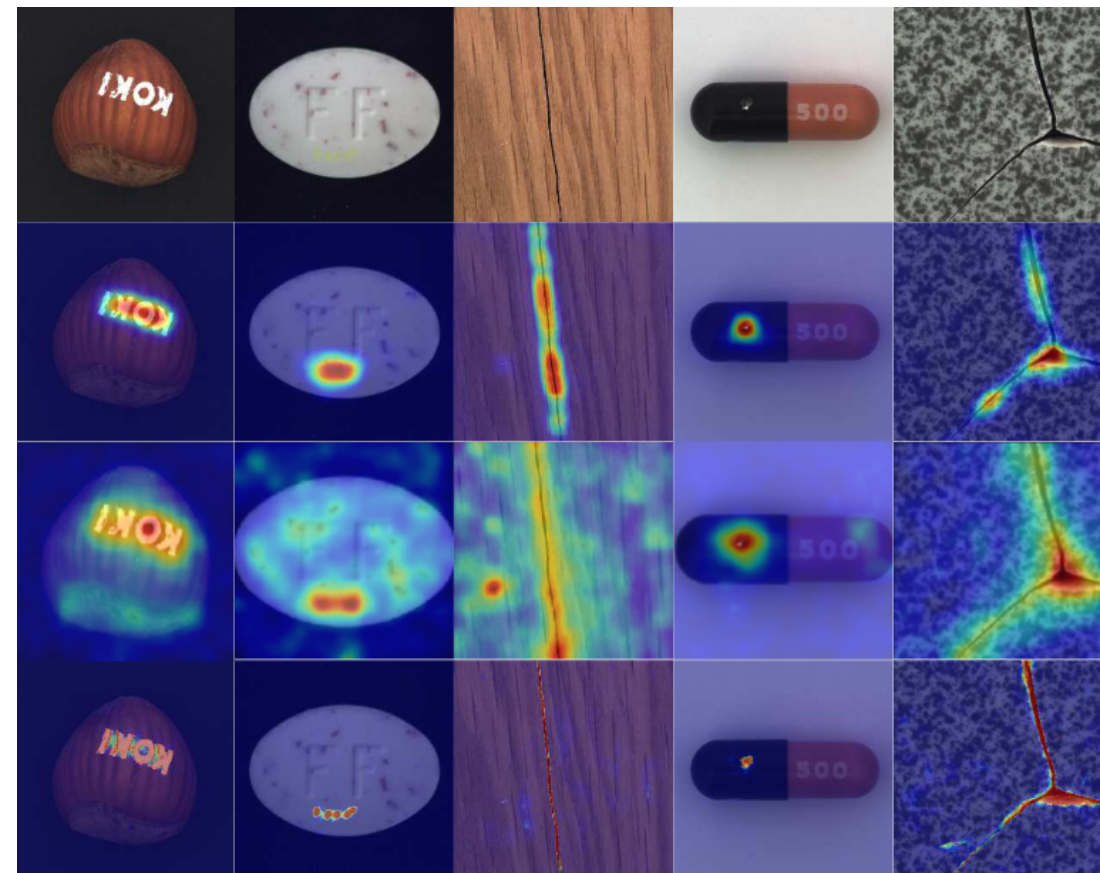


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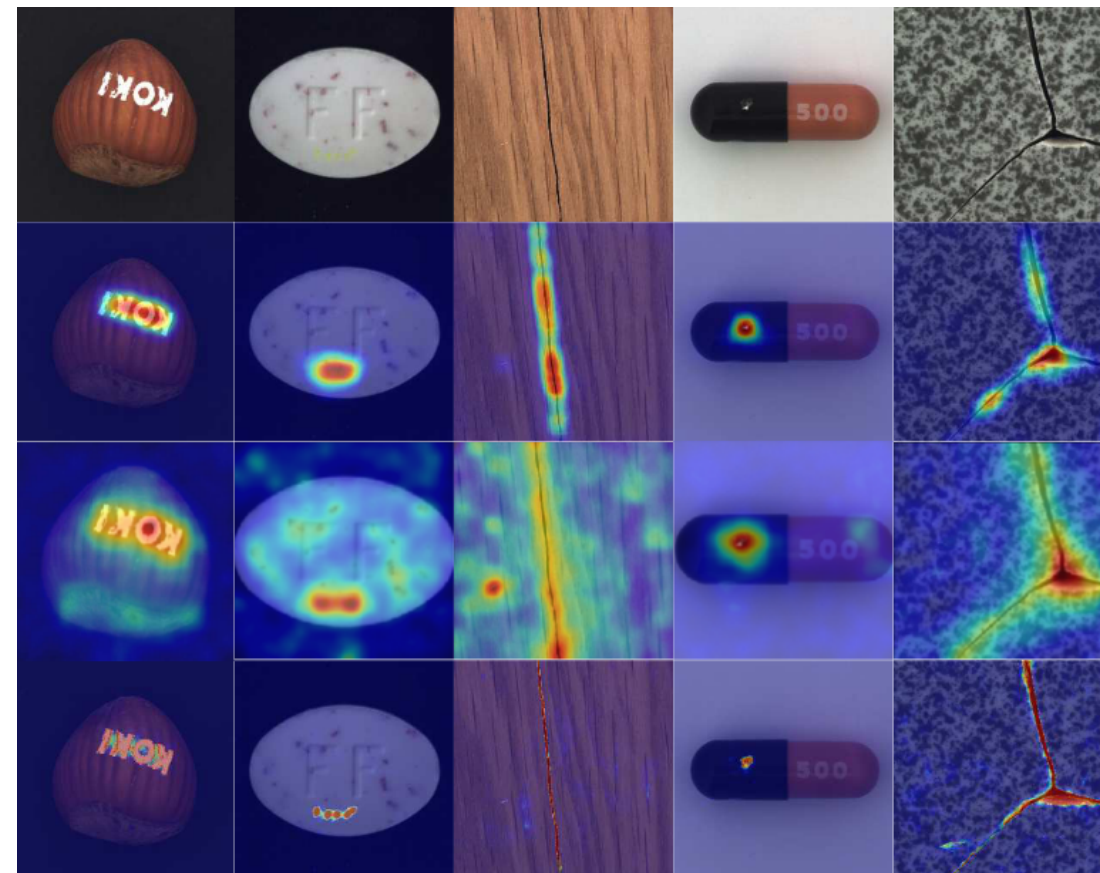
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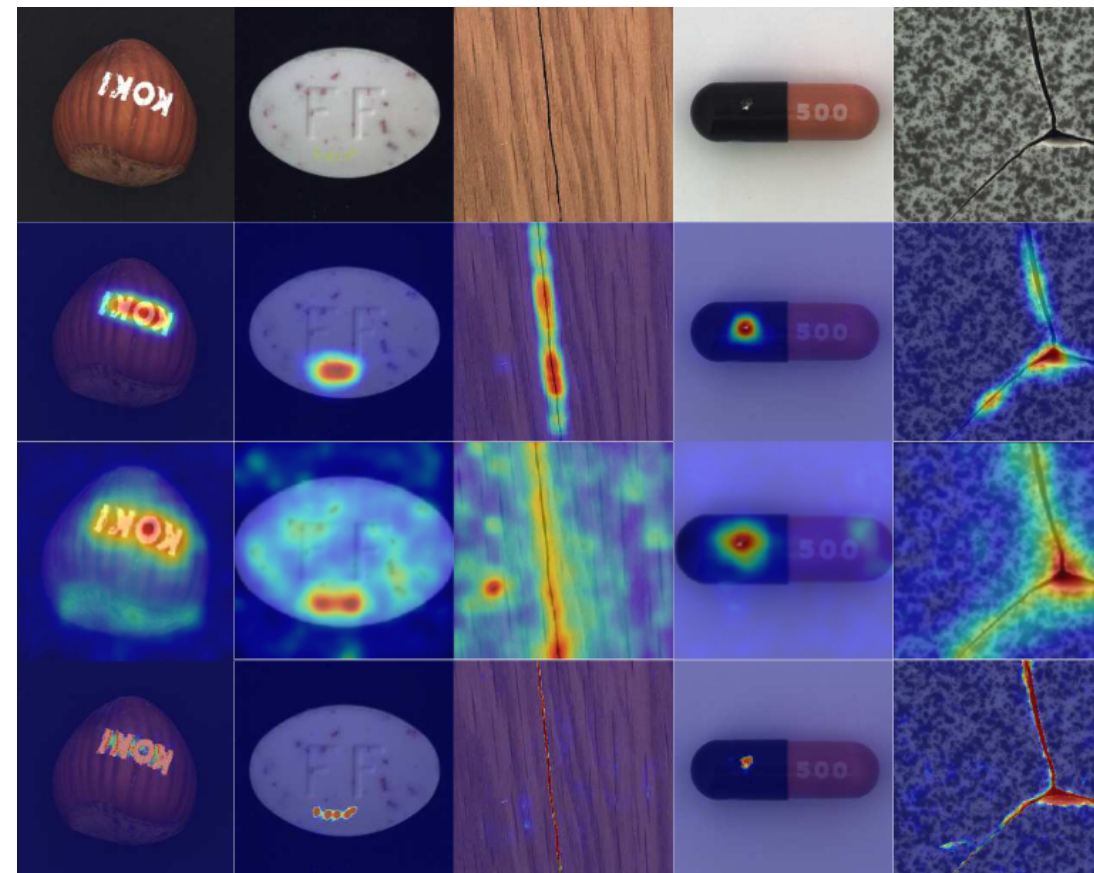




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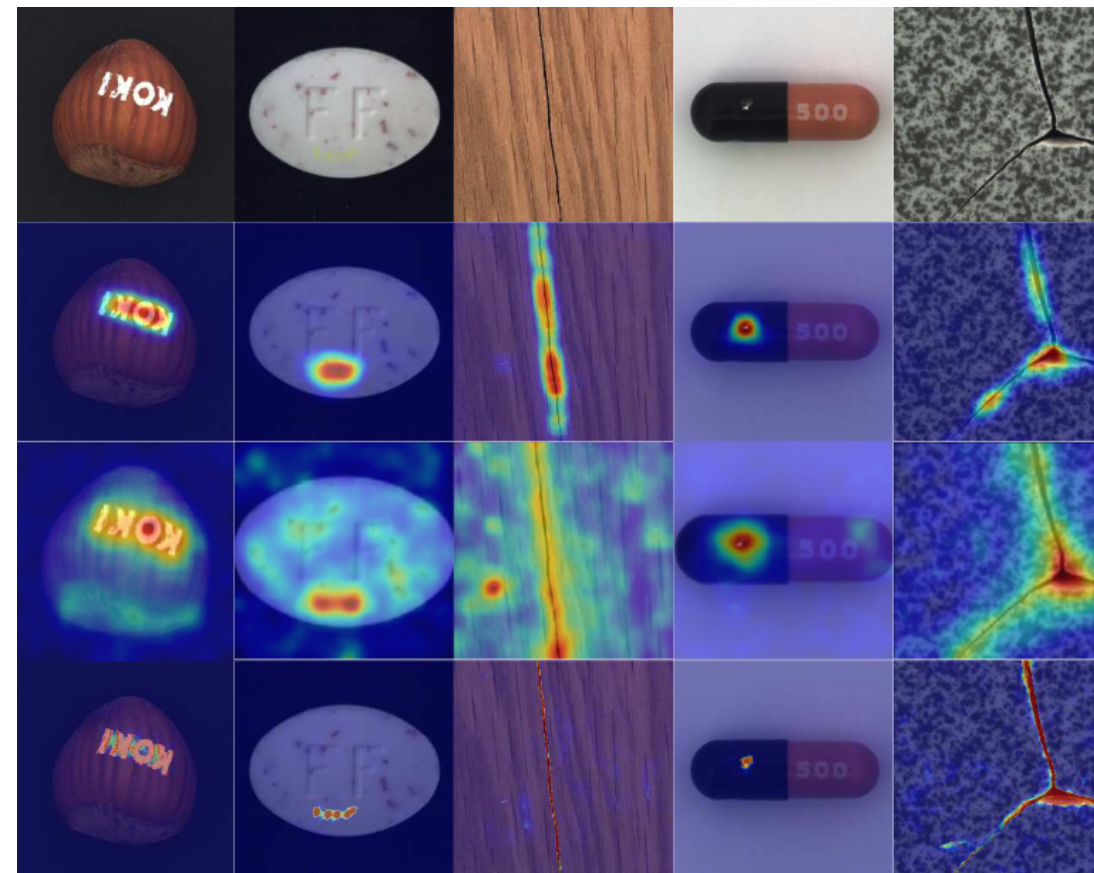
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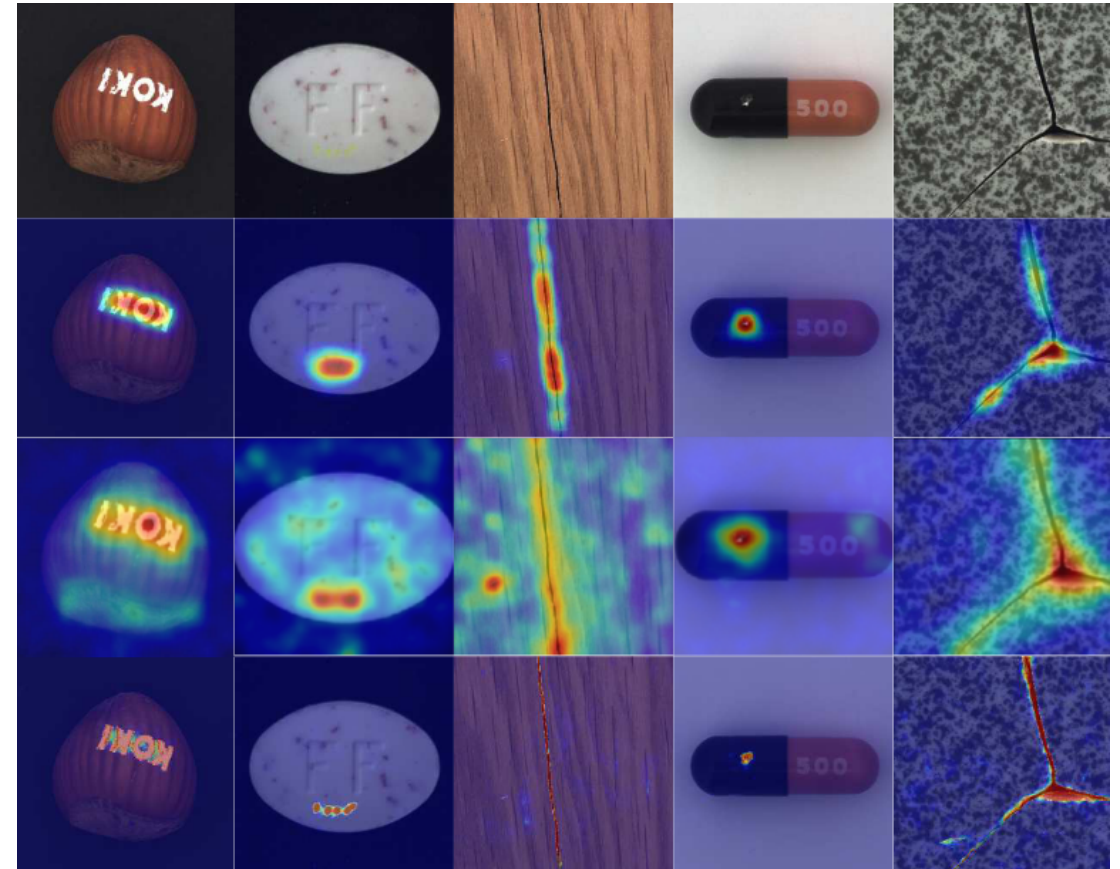
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  - Estimate the size of the impacted area
  - Distinguish between cracks, corrosion, or just paint
- We don't need expensive hand-labeled data
- We only need just-out-of-distribution patterns for anomaly generation



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Figure: Loss smoothing by weighting the last batch of each epoch.



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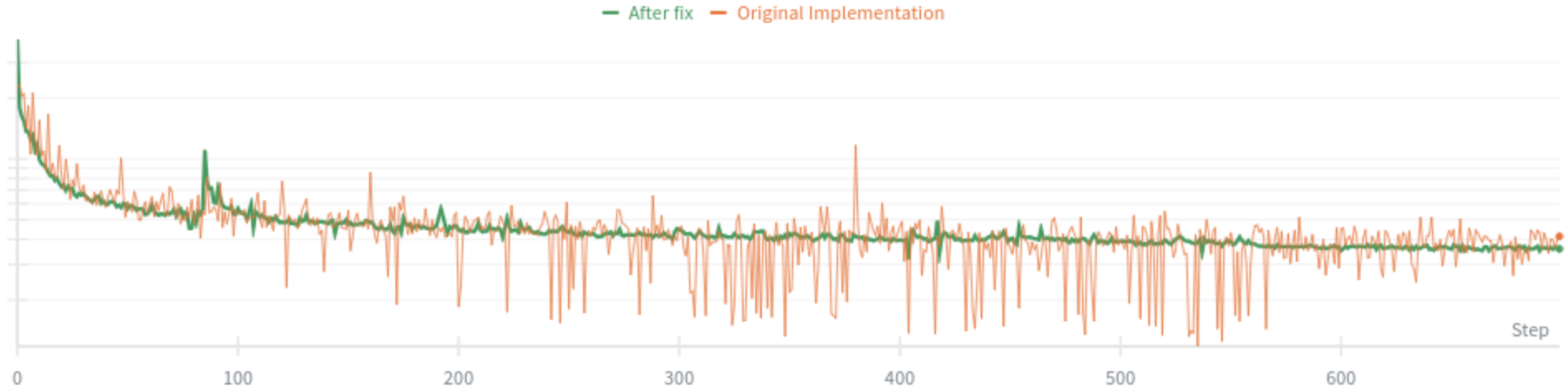


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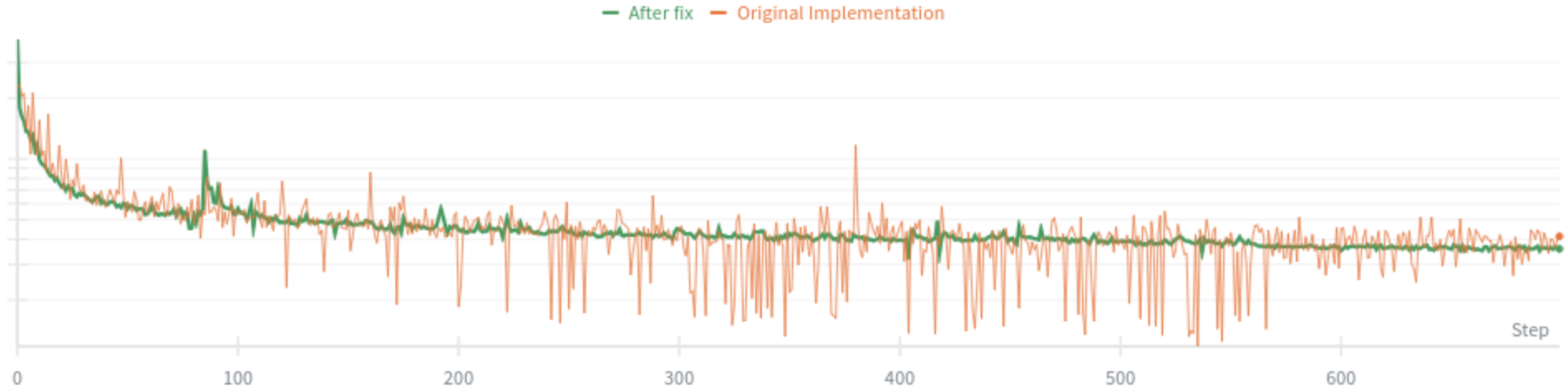


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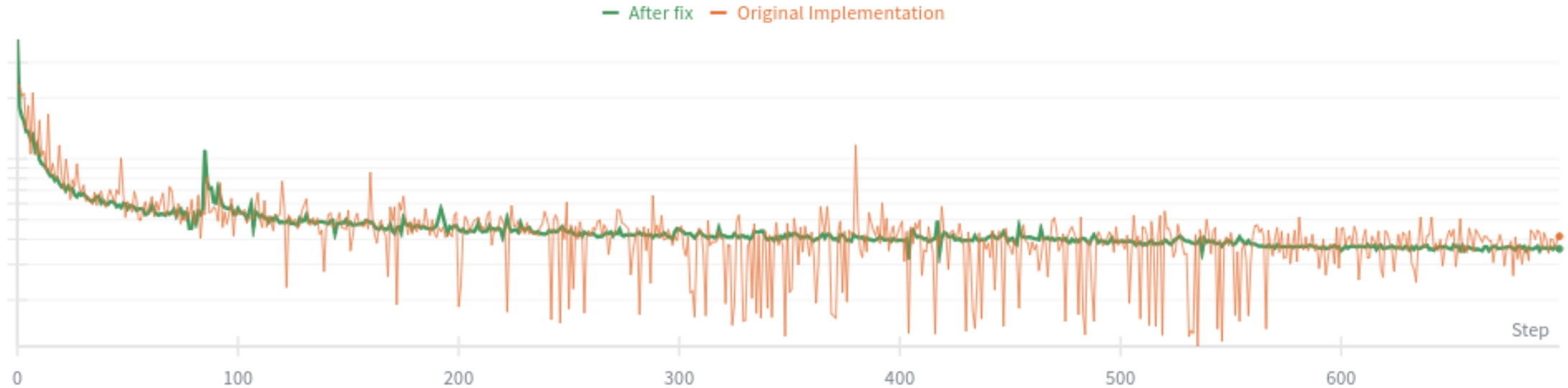


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- ✓ Well readable code
- ✓ Download script for MVTec and DTD datasets

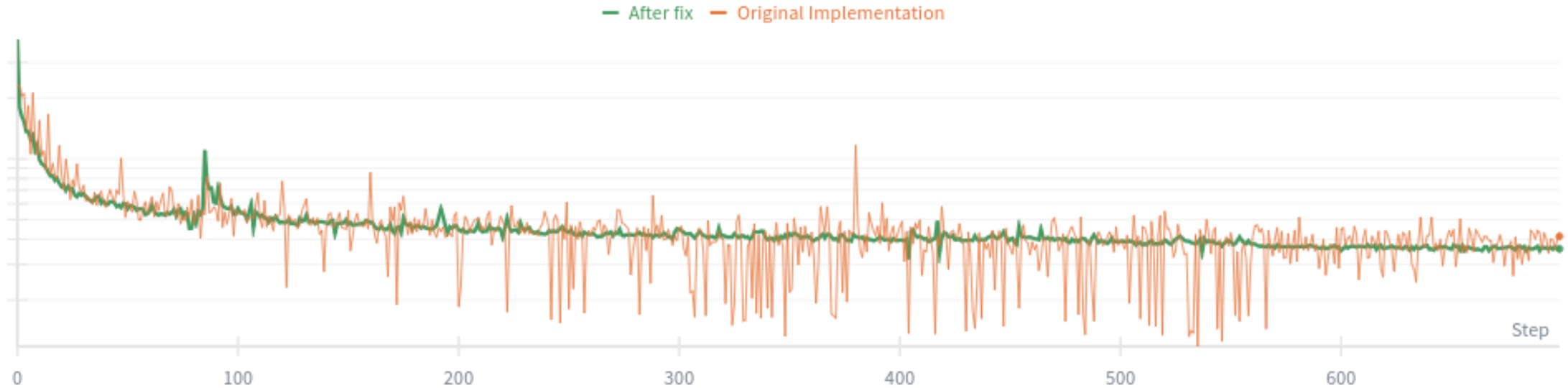


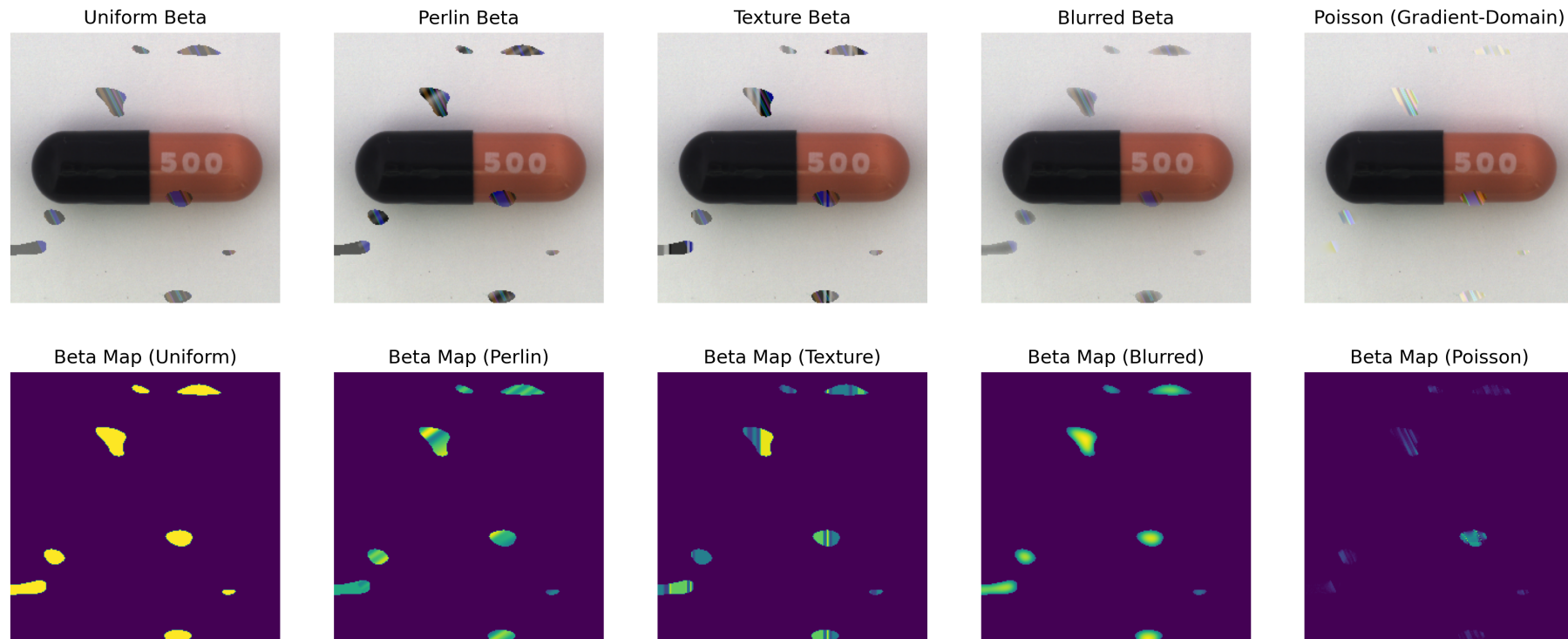
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**Figure:** Different blending methods (from left to right): uniform  $\beta$ , Perlin- $\beta$ , texture- $\beta$ , blurred  $\beta$ , Poisson Image Interpolation.

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| DRÆM-Blurred | 96.4%        | 96.0%        | 65.8%        |
| DRÆM-Texture | 98.0%        | 97.2%        | 69.1%        |
| DRÆM-Perlin  | 98.2%        | 97.3%        | 69.5%        |
| DRÆM-Poisson | <u>99.1%</u> | <u>97.8%</u> | <u>71.6%</u> |

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| DRÆM-Perlin         | 98.2%        | 97.3%        | 69.5%        |
| DRÆM-Poisson        | <u>99.1%</u> | <u>97.8%</u> | <u>71.6%</u> |
| DRÆM-Poisson+Perlin | <b>99.2%</b> | <b>98.0%</b> | <b>73.6%</b> |

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**Thank you for your attention.**

The background of the slide is a solid blue color. In the lower half, there is a decorative graphic consisting of numerous thin, concentric, curved lines that create a sense of motion or a ripple effect, starting from the bottom center and curving outwards towards the sides.